CSAT Hard Revision Folder 3: Arrangements, coding, counting, relations, syllogisms and analytical constraints

ANSWER KEY + SOLUTIONS | 40 questions | Source: original synthesis

Scoring: +2.5 for correct, –5/6 for incorrect, 0 for blank. Check the key only after completing the answer-free paper.

Q	Key	Category
1	C	Comprehension
2	A	Comprehension
3	B	Comprehension
4	A	Comprehension
5	C	Comprehension
6	A	Comprehension
7	D	Comprehension
8	B	Comprehension
9	A	Comprehension
10	B	Comprehension
11	B	Numeracy / DI
12	A	Numeracy / DI
13	B	Numeracy / DI
14	D	Numeracy / DI
15	B	Numeracy / DI
16	C	Numeracy / DI
17	A	Numeracy / DI
18	C	Numeracy / DI
19	B	Numeracy / DI
20	D	Numeracy / DI
21	D	Logical reasoning
22	B	Logical reasoning
23	B	Logical reasoning
24	B	Logical reasoning
25	A	Logical reasoning
26	A	Logical reasoning
27	D	Logical reasoning
28	A	Logical reasoning
29	D	Logical reasoning
30	D	Logical reasoning
31	A	Decision-making / problem solving
32	C	Decision-making / problem solving
33	B	Decision-making / problem solving
34	C	Decision-making / problem solving

Q	Key	Category
35	A	Decision-making / problem solving
36	B	Decision-making / problem solving
37	C	Decision-making / problem solving
38	B	Decision-making / problem solving
39	A	Decision-making / problem solving
40	C	Decision-making / problem solving

DETAILED SOLUTIONS

Q1 — C Comprehension
The passage links the issue to incentives, evidence and feedback; the alternatives are absolute or narrowly framed.
Provenance: Original synthesis

Q2 — A Comprehension
The passage distinguishes a desirable objective from the quality of its feedback loop; it rejects universal claims.
Provenance: Original synthesis

Q3 — B Comprehension
The argument depends on learning through feedback; it does not require perfect information or reject all central rules.
Provenance: Original synthesis

Q4 — A Comprehension
The wording weighs trade-offs and improvement mechanisms rather than celebrating, rejecting or attacking them.
Provenance: Original synthesis

Q5 — C Comprehension
The passage links the issue to incentives, evidence and feedback; the alternatives are absolute or narrowly framed.
Provenance: Original synthesis

Q6 — A Comprehension
The passage distinguishes a desirable objective from the quality of its feedback loop; it rejects universal claims.
Provenance: Original synthesis

Q7 — D Comprehension
The argument depends on learning through feedback; it does not require perfect information or reject all central rules.
Provenance: Original synthesis

Q8 — B Comprehension
The wording weighs trade-offs and improvement mechanisms rather than celebrating, rejecting or attacking them.
Provenance: Original synthesis

Q9 — A Comprehension
The passage links the issue to incentives, evidence and feedback; the alternatives are absolute or narrowly framed.
Provenance: Original synthesis

Q10 — B Comprehension
The passage distinguishes a desirable objective from the quality of its feedback loop; it rejects universal claims.
Provenance: Original synthesis

Q11 — B Numeracy / DI
Compute powers modulo 12; cyclic reduction gives remainder 5.
Provenance: Original synthesis

Q12 — A Numeracy / DI
Count multiples of 7 and 12, then remove twice the common multiples: the exclusive count is 18.
Provenance: Original synthesis

Q13 — B Numeracy / DI
Use a weighted average: (8×21+20)/(8+1) = 20.89%.
Provenance: Original synthesis

Q14 — D Numeracy / DI
Combined rate = 1/16+1/27+1/39; reciprocal = 7.99 days.
Provenance: Original synthesis

Q15 — B Numeracy / DI
Change in average = (new−old)/23; new average = 63.17.
Provenance: Original synthesis

Q16 — C Numeracy / DI
Multiply successive factors: 1.0×(1+16/100)×(1−21/100)×(1+26/100); net = 15.47%.
Provenance: Original synthesis

Q17 — A Numeracy / DI
Removed area = 4×3^2; remaining area = $19\times11-4\times3^2 = 173$ m².
Provenance: Original synthesis

Q18 — C Numeracy / DI
Weighted total = 158; 158 divided by 17 leaves remainder 5.
Provenance: Original synthesis

Q19 — B Numeracy / DI
lcm(10,10) = 10; enumerate its multiples in the interval to obtain 3.
Provenance: Original synthesis

Q20 — D Numeracy / DI
The increments alternate 3, 4, 3, 4; the next increment is 3, giving 21.
Provenance: Original synthesis

Q21 — D Logical reasoning
Enumerating all valid arrangements gives possible fifth-position occupants ['A', 'B', 'D', 'E', 'F', 'G']; C never appears there.
Provenance: Original synthesis

Q22 — B Logical reasoning
Testing all rankings gives possible third-place candidates ['P', 'Q', 'R', 'S', 'T']; U violates the partial-order constraints in every valid ranking.
Provenance: Original synthesis

Q23 — B Logical reasoning
(7×5+3) mod 26 = 12.
Provenance: Original synthesis

Q24 — B Logical reasoning
Y contains X plus Y-only members: 4+6 = 10.
Provenance: Original synthesis

Q25 — A Logical reasoning
Displacement = sqrt(6^2+6^2) = 8.49 km.
Provenance: Original synthesis

Q26 — A Logical reasoning
N is P's father; M is N's sister, so M is P's paternal aunt.
Provenance: Original synthesis

Q27 — D Logical reasoning
Assume exactly one of A, B and C is guilty and exactly one statement is true. A says B is guilty; B says A and C are innocent; C says B is innocent. Testing A, B and C in turn leaves only B guilty: A and C are then false, while B is true.
Provenance: Original synthesis

Q28 — A Logical reasoning
All selections minus those containing both specified proposals: C(7,3)–C(5,1) = 30.
Provenance: Original synthesis

Q29 — D Logical reasoning
Points = 3×4+3 = 15; losses add zero.
Provenance: Original synthesis

Q30 — D Logical reasoning
The first post has 7 choices and the second has 6; ordered pairs therefore total 7×6 = 42.
Provenance: Original synthesis

Q31 — A Decision-making / problem solving
Triangulation tests measurement and lived experience before a consequential administrative action.
Provenance: Original synthesis

Q32 — C Decision-making / problem solving
Responsible risk communication is candid about uncertainty while giving a clear time-bound action.
Provenance: Original synthesis

Q33 — B Decision-making / problem solving
A transparent composite rule recognises scale and severity while making trade-offs reviewable.
Provenance: Original synthesis

Q34 — C Decision-making / problem solving
The indicator must track intended outcomes and be protected against gaming through verification.
Provenance: Original synthesis

Q35 — A Decision-making / problem solving
The process respects the association's input without mistaking it for complete representation.
Provenance: Original synthesis

Q36 — B Decision-making / problem solving
Disaggregated evidence supports adaptation rather than premature scaling, cancellation or concealment.
Provenance: Original synthesis

Q37 — C Decision-making / problem solving
Complaint counts need accessibility and independent-feedback checks before they can indicate improvement.
Provenance: Original synthesis

Q38 — B Decision-making / problem solving
A documented risk and change-control assessment protects safety without making procedure an empty ritual.
Provenance: Original synthesis

Q39 — A Decision-making / problem solving
Transparent deduplication plus appeal and verification addresses both leakage and wrongful exclusion.
Provenance: Original synthesis

Q40 — C Decision-making / problem solving
Equal application of a published rule protects fairness while allowing genuinely defined hardship relief.
Provenance: Original synthesis